INCH-POUND

MS25271K
26 August 2020
SUPERSEDING
MS25271J
w/AMENDMENT 1
15 December 2004

DETAIL SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 10 AMPERES, 4 PDT, TYPE I, SOLDER HOOK, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 15 NOVEMBER 2002.
NO SUPERSEDING SPECIFICATION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

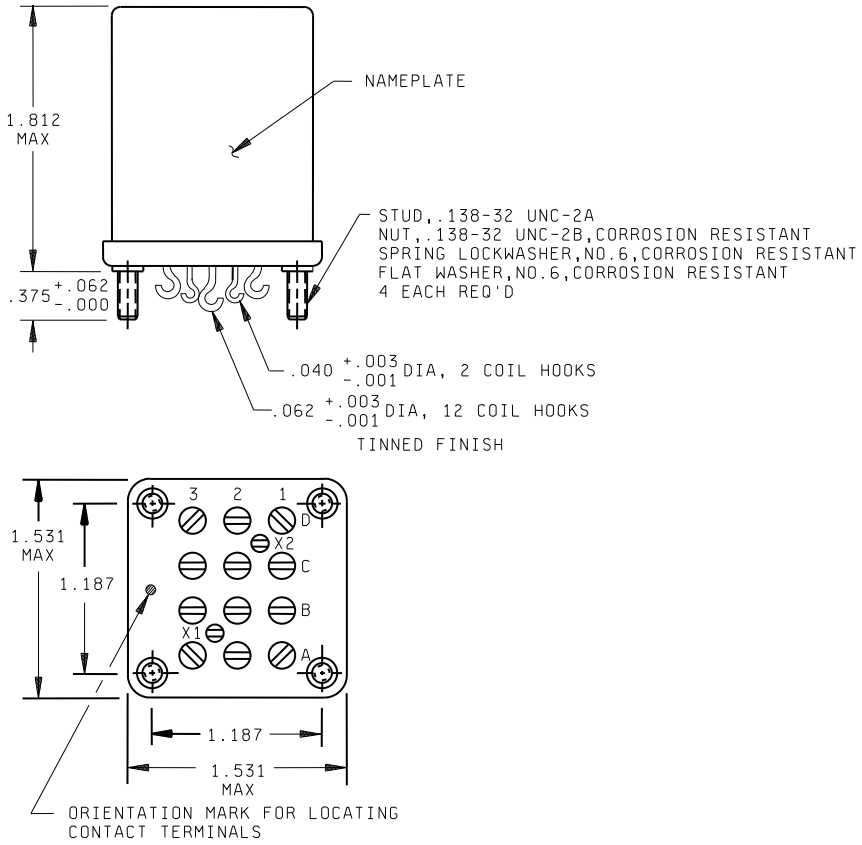
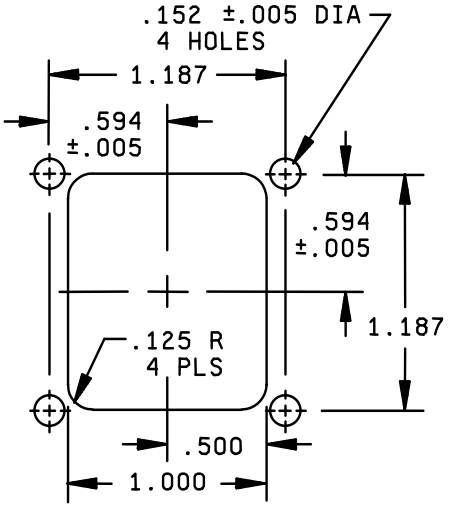
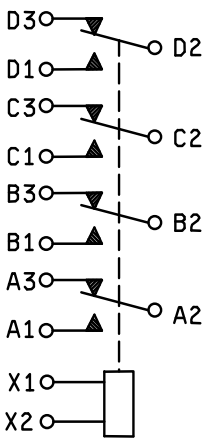


FIGURE 1. Dimensions and configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:07Z
Check the source to verify that this is the current version before use.



MOUNTING HOLE LAYOUT



CIRCUIT DIAGRAM

Inches	mm	Inches	mm
.001	0.03	.152	3.86
.003	0.08	.375	9.53
.005	0.13	.500	12.70
.010	0.25	.594	15.09
.040	1.02	1.000	25.40
.062	1.57	1.187	30.15
.125	3.18	1.531	38.89
.138	3.51	1.812	46.02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay header provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
5. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
6. Referenced Government documents of the issue listed in Assist Online (https://assist.dla.mil/) or Assist Quick Search (https://quicksearch.dla.mil/) specified in the solicitation form a part of this specification to the extent specified herein.

FIGURE 1. Dimensions and configurations - Continued.

TABLE I. Dash numbers and characteristics.

Dash number MS25271-	Type	Coil	Terminal type	Mounting	Max weight in pounds
D1	I	dc	Solder hook	Stud	.45
A1	I	ac	Solder hook	Stud	.45

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:07Z
Check the source to verify that this is the current version before use.

TABLE II. Operating characteristics.

PIN MS25271-	Coil data											Time - milliseconds max					
	Coil	Rated			Max		Max pick-up voltage			Drop out voltage 2/	Hold voltage 2/	Operate 3/	Release 4/	Contact Bounce			
														Main		Aux	
		Volts 1/	Freq Hz	Res Ω ±10%	Volts	Amp	Normal 2/	High temp. test	Cont current test					NO	NC	NO	NC
D1	X1, X2	28	dc	110	29	0.350	18	19.5	22.5	1.5	7.0	20	20	2	2	N/A	N/A
A1	X1, X2	115	400 5/	N/A	122	0.073	90	95	103	5.0	30	25	50	2	2	N/A	N/A

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ MS25271-A1 may be used on 60 Hz if maximum ambient temperature is +85°C (coil current shall be 0.077 ampere maximum).

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			6	4			6	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Mixed loads	Applicable in accordance with MIL-PRF-6106.													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Environmental characteristics.

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 g's
Duration	11 ms
Max duration contact opening	10 μs

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:07Z
Check the source to verify that this is the current version before use.

Vibration - sinusoidal (see chart below)	
G-level	10 g's
Frequency range	5-1,500 Hz
Vibration – random	
Applicable specification	N/A
Power spectral density	N/A
RMS G min	N/A
Frequency range	N/A
Curve	N/A
High shock	N/A
Acceleration	15 g's

Electrical characteristics.

Insulation resistance, initial	100 megohms.
After life or environmental tests	50 megohms.

Dielectric strength (sea level).

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude).

	80,000 ft
Coil to case	250 V rms
Aux contacts	
All other points	350 V rms

Max contact drop initial	0.150 volt.
After life test	0.175 volt.
Overload current	20 amperes.
Rupture current	25 amperes.
Duty rating	Continuous.
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays).

Qualification by similarity: See MIL-PRF-6106.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:07Z
Check the source to verify that this is the current version before use.

Conformance inspection.

Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes, and shall impose additional testing requirements as necessary.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Review activities:
Navy - EC

Preparing activity:
DLA - CC

(Project 5945-2020-046)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:07Z
Check the source to verify that this is the current version before use.